Introduction, Exponential Smoothing & Forecast Evaluation

ECON 8310: Business Forecasting — Lecture 1

Department of Economics

University of Nebraska at Omaha

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Lecture Outline

1 Why Forecasting Matters

2 Exponential Smoothing

3 Forecast Evaluation

4 Course Roadmap

Why Forecasting Matters

Every business decision made today depends on expectations about tomorrow.

Forecasting Is Everywhere in Business

Domain	The question someone is actually asking
Operations & supply chain	How much inventory should Walmart order for Black Friday? How many nurses does a hospital need next Tuesday night?
Finance & risk	Will credit card defaults rise next quarter? What will the Fed Funds rate be in six months?
Marketing & strategy	How many units will this new product sell in year one? What is the lifetime value of this customer cohort?
Macro & public policy	Will GDP grow by 2% or 3% next year? What will unemployment be in Q4?

Every one of these is a *decision* waiting on a number. **A decision is only as good as the forecast it rests on.**

The Basic Setup

Time Series

A **time series** is an ordered sequence of observations $\{y_1, y_2, \ldots, y_T\}$ where t indexes time.

Notation for the whole course:

- y_t — observed value at time t
- h — **forecast horizon**: harder as h grows, as uncertainty compounds
- $\hat{y}_{t+h|t}$ — forecast of y_{t+h} made using information at time t
- $e_t = y_t - \hat{y}_{t|t-1}$ — **forecast error**

Under squared-error loss the best forecast is the **conditional expectation**, $\hat{y}_{t+h|t} = \mathbb{E}[y_{t+h} \mid \mathcal{F}_t]$, where $\mathcal{F}_t$ is everything known at time t.

Components of a Time Series

Any series can be decomposed into four parts (additive form): $y_t = T_t + S_t + C_t + I_t$.

- T_t **Trend** — long-run direction
- S_t **Seasonality** — regular, calendar-driven pattern
- C_t **Cycle** — medium-run business-cycle swings
- I_t **Irregular** — random shocks

Which parts matter depends on the series: retail sales are trend + seasonality, quarterly GDP is trend + cycle, daily stock returns are mostly irregular.

Ignoring seasonality causes large, systematic errors. A non-seasonal model on monthly retail data will consistently under-forecast December and over-forecast January — every single year.

Always Beat a Benchmark First

Before presenting any forecast model, check that it beats a **naive benchmark**. A model that cannot beat the baseline has zero value.

Benchmark	Forecast	Use when
Naïve	$\hat{y}_{t+h\|t} = y_t$	Asset prices, random walks
Seasonal naïve	$\hat{y}_{t+h\|t} = y_{t+h-m}$	Any series with a season
Mean	$\hat{y}_{t+h\|t} = \bar{y}_t$	No trend, no season
Drift	$\hat{y}_{t+h\|t} = y_t + h\hat{c}$	Trend, no season

This is not a formality. In the M4 Competition many machine learning models *failed* to beat simple exponential smoothing (Makridakis et al. 2020).

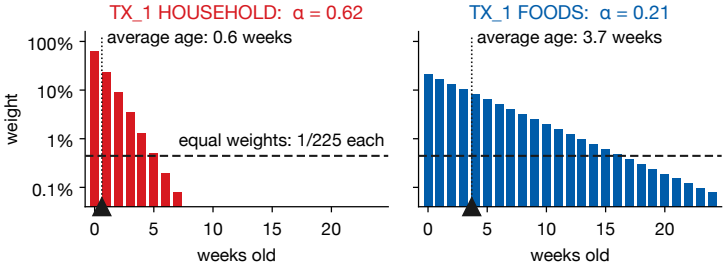
Exponential Smoothing

Recent observations should carry more weight than old ones.

The Problem with Equal Weights

Most classical methods weight every observation equally — but what if the world **changes**? After a store renovation, equal weights keep forecasting the old store for months.

Exponential weighting: an observation k weeks old gets weight $\alpha(1-\alpha)^k$. High α = short memory; low α = long memory:



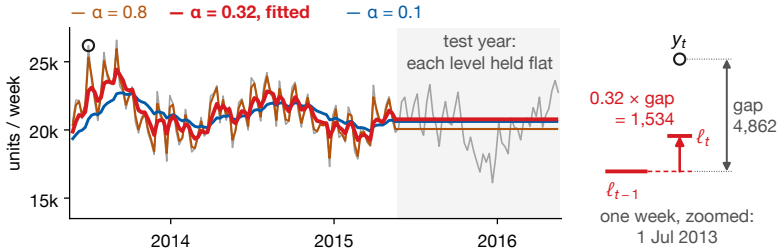
Fitted α on two aisles of one store. Log scale: each week back is the same step down, $\times(1-\alpha)$.

Simple Exponential Smoothing (SES)

SES Update Equation

The **level** ℓ_t closes the fraction α of each week's gap ($0 < \alpha \leq 1$); the forecast is flat.

$$\ell_t = \alpha\, y_t + (1-\alpha)\, \ell_{t-1} = \ell_{t-1} + \alpha\,(y_t - \ell_{t-1}), \qquad \hat{y}_{T+h|T} = \ell_T$$



Large α chases every wiggle; small α is smooth but slow. SES **nests both benchmarks**: $\alpha = 1$ is naïve, $\alpha \to 0$ the historical mean. *No trend, no season.*

Holt's Method: Adding Trend

SES is *flat*. For a trending series, track the level ℓ_t and the slope b_t:

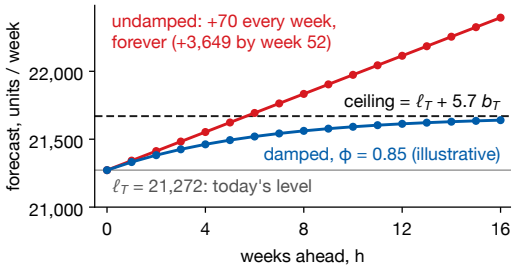
Holt's Linear Exponential Smoothing (Holt 2004)

Level: $\ell_t = \alpha\, y_t + (1-\alpha)\,(\ell_{t-1} + b_{t-1})$

Trend: $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)\, b_{t-1}$

Forecast: $\hat{y}_{T+h|T} = \ell_T + h\, b_T$

Straight lines get unrealistic far out. For $h > 6$, the **damped trend** multiplies each week's rise by ϕ $(0 < \phi < 1)$, so the forecast levels off under $\ell_T + b_T\, \phi/(1-\phi)$.



Holt-Winters: Level, Trend, and Seasonality

Holt-Winters Additive Method (Winters 1960)

$$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1})$$
$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)\, b_{t-1}$$
$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1-\gamma)\, s_{t-m}$$
$$\hat{y}_{T+h|T} = \ell_T + h\, b_T + s_{T+h-m} \qquad (m = \text{seasonal period})$$

Additive seasonality:

- Swings constant in size: the same $\pm$ amount every December

Multiplicative seasonality:

- Swings grow with the level: retail sales, tourism

US retail's December spike grows as overall sales grow: **multiplicative**.

Needs at least $2m$ observations to initialize ($m = 12$ for monthly data).

The ETS Framework: A Unified View

Hyndman and Athanasopoulos (2021) unify every exponential smoothing method as a **state space model**: Error $\times$ Trend $\times$ Seasonal.

Component	Options	Meaning
Error (E)	A, M	Additive / Multiplicative
Trend (T)	N, A, A_d	None / Additive / Damped
Seasonal (S)	N, A, M	None / Additive / Multiplicative

That is 18 combinations; three are numerically unstable, which leaves 15. Ours are among them: **ETS(A,N,N)** is SES, **ETS(A,A,N)** is Holt linear, **ETS(A,A_d,N)** is Holt damped, **ETS(A,A,A)** is Holt-Winters additive, and **ETS(M,A,M)** is Holt-Winters multiplicative.

Automatic selection: fit all 15 and pick the minimum AIC. In Python, `statsmodels.tsa.exponential_smoothing.ets.ETSModel`.

Forecast Evaluation

Forecasting without evaluation is not science — it is storytelling.

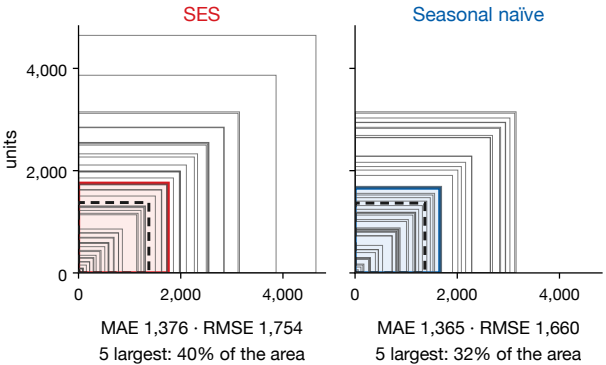
How Do We Measure Forecast Accuracy?

Draw each out-of-sample error $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$ as a square. CA_1 FOODS, the 52 test weeks:

$$\mathrm{MAE} = \frac{1}{H}\sum_{h=1}^{H} |e_{T+h}|$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{H}\sum_{h=1}^{H} e_{T+h}^2}$$

MAE is the average **side** (dashed).
RMSE is the side of the average **square** (filled), so $\mathrm{RMSE} \geq \mathrm{MAE}$.



A tie on MAE, not on RMSE: five big misses carry 40% of SES's squared error. When big misses are costly — a stockout — prefer RMSE.

Comparing Across Series: MAPE and MASE

RMSE and MAE come in the units of y_t, so they compare models on the *same* series. To compare *across* series, scale the errors:

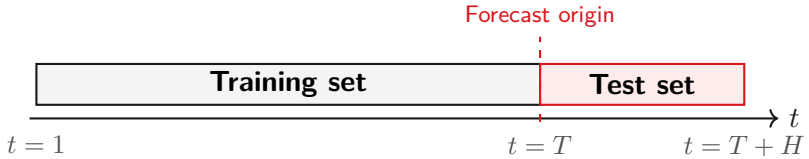
$$\text{MAPE} = \frac{100\%}{H}\sum_{h=1}^{H}\left|\frac{e_{T+h}}{y_{T+h}}\right| \qquad \text{MASE} = \frac{\text{MAE}}{\frac{1}{T-m}\sum_{t=m+1}^{T}|y_t - y_{t-m}|}$$

The MASE denominator is seasonal naïve's average miss **on the training data**. MASE < 1: smaller errors than that; MASE > 1: worse. It is the standard in the M-Competitions (Makridakis et al. 2020).

MAPE breaks down when y_t is near zero; MASE does not.

The Golden Rule: Always Evaluate Out-of-Sample

Never evaluate a forecast model on the data used to fit it. In-sample fit measures how well the model memorized the past, not how well it predicts the future.



Never randomly shuffle time series observations. Use `TimeSeriesSplit` in scikit-learn, never `KFold`. Shuffling lets future data train the model — data leakage that makes results look excellent and forecast badly.

Course Roadmap

What we will cover, and how it fits together.

ECON 8310: Four-Part Structure

Part	Lectures
I. Time series	1. Intro, ETS & evaluation ← today 2. ARIMA & VAR 3. Generalized additive models
II. Tree-based	4. Decision trees 5. Tree ensembles: random forests & boosted trees 6. Regularization & model selection
III. Deep learning	7. Neural networks (PyTorch) 8. CNN architectures 9. RNNs, LSTMs & Transformers
IV. Bayesian	10. Foundations 11. Time series & hierarchical models 12. Bayesian linear regression

Python throughout: statsmodels, scikit-learn, xgboost, PyTorch, prophet, pyGAM, pymc.

Tools and What You Will Be Able to Do

The Python stack: `statsmodels` (ETS, ARIMA, VAR) · `scikit-learn` (models, pipelines, cross-validation) · `xgboost` (gradient boosting) · `torch` (neural networks) · `prophet` and `pygam` (GAMs) · `pymc` (Bayesian models).

By the end of the semester you will be able to:

- ✓ Choose the right method for a given business problem
- ✓ Implement and tune it in Python
- ✓ Evaluate it rigorously out-of-sample
- ✓ Interpret the result for a non-technical stakeholder
- ✓ Use AI tools effectively to accelerate the analysis
- ✓ Avoid the most common forecasting mistakes

Lecture 1: Key Takeaways

1. Under squared loss the best forecast is the **conditional expectation**.
2. **Always beat a benchmark first:** naïve, seasonal naïve, mean, drift.
3. Exponential smoothing weights the past **geometrically**; α sets how fast it forgets.
4. SES $\rightarrow$ Holt ($+$ trend) $\rightarrow$ Holt-Winters ($+$ season): all **ETS** models, chosen by AIC.
5. **Evaluate out-of-sample, never shuffled.** RMSE punishes big misses; MASE compares across series.

Next: ARIMA & VAR — a regression-based approach to time series with rich autocorrelation structure.

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- Hyndman, Rob J. and George Athanasopoulos (2021). *Forecasting: Principles and Practice*. 3rd. Open access. Melbourne, Australia: OTexts. URL: https://otexts.com/fpp3/.
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